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Re: Request for Information (RFI) on Development of an AI Action Plan - 3/15/25

Summary

Brightband strongly supports OSTP's efforts to plan America's leadership in AI. Weather forecasting is critical for national security, a stronger economy, and more competitive military. Europe's weather forecasting model has become the most accurate global standard. AI for Weather Prediction (AIWP) is the biggest step forward in forecasting since satellites and supercomputers in the 1960s. China¹ is planning an operational AIWP forecasting system and Europe already has one. The USA does not yet have an operational AIWP model, and needs one soon to lead in AI and Weather.

To ensure America achieves global leadership in AI for Weather Prediction the administration should support an "AIWP Moonshot":

1. NOAA should deliver an operational AIWP model for global medium-term forecasting, with performance competitive to ECMWF's AIFS, by July 4, 2026.
2. NOAA should source the latest AIWP/data technology from private industry partners via acquisition, licensing, or co-development.
3. NOAA, NASA, DoD, DoE, and other agencies should coordinate and leverage the USA's unparalleled Earth Observation capabilities to provide the world's leading real-time datastream of weather observations for AIWP modeling.

About Brightband

Brightband is a leading AI weather forecasting model developer, an American startup, founded by a team of serial entrepreneurs in the AI and weather space. Brightband is the first private company to develop a global, operational weather modeling system forecasting directly from weather observations. Brightband is helping NOAA accelerate AIWP by re-processing weather observation datasets to be ready for the cloud and AI.

Opportunity

USA GDP fluctuates by >3% annually due to weather (almost \$1Trillion)². 1/3 of the USA economy is weather dependent³. Extreme weather caused \$396Billion of damage in the

¹ Bloomberg, [China's AI Weather Forecasting Goals Face National Security Test](#), 2025

² [US economic sensitivity to weather variability](#) (Lazo et al, 2011)

³ Table 2, [Opportunities and priorities in a new era for weather and climate services](#) (Dutton, 2002)

USA in 2017⁴. Weather forecasting is crucial for the military, national security, farmers, supply chain operations, space leadership, and saving lives amid disasters.

As a global superpower fielding a military with global reach, the United States relies on having the fastest and most reliable weather forecasts as a critical pillar of safeguarding its national security. Our military history demonstrates weather forecasting's strategic importance. Fog helped conceal Washington's troops during the evacuation of Brooklyn and a nor'easter prevented the Royal Navy from entering the East River. President Eisenhower's decision to delay D-Day due to bad weather saved thousands of Allied lives and helped secure victory in World War II. Advanced forecasting of the unpredictable weather in the mountains helped save countless lives in Afghanistan. It was critical to predict sandstorms and heat in the Middle East. Weather significantly impacts modern military operations, affecting every aspect from planning and logistics to execution and outcomes, and can be leveraged for tactical advantage, such as influencing enemy behavior or delaying/canceling operations. From making logistics and operations resilient despite tropical cyclones in the Pacific to pinpointing when cloud decks will break for aerial reconnaissance and support, sustained improvement in weather forecasting through innovation and AI leadership is key to American military competitiveness and national security. Today the US Air Force – who are the primary provider of weather forecast data in support of the American warfighter – has built its current NWP capabilities on top of the UK Met Office's modeling infrastructure, the Unified Model.

Weather is also key to promoting America's prosperity and commercial competitiveness. It is the most commonly cited external factor affecting changes in financial results, according to filed reports. The United States has 900 million acres of farm land that face a unique variety of weather extremes. Weather forecasting is key to optimizing agricultural productivity. Farmers need to know when to buy seeds, plant, fertilize, irrigate, and harvest, informed by predictions of sunshine, rain, frost, wind etc. to maximize efficiency. Energy infrastructure safety and efficiency are dependent on weather. Winds or temperature extremes can cause infrastructure to fail, wildfires or blackouts. Energy efficiency planning is largely informed by weather. Solar and wind generation depend on weather predictions, and energy demand varies with the heating/cooling required based on temperature changes. America's supply chain and transportation safety and efficiency rely on aviation plotting safe trajectories through predicted clouds, ice, wind and weather hazards, trucks avoiding icy roads, and ships steering clear of big waves. America's retailers rely on weather forecasting to know whether to sell hot apple pie or ice cream. For timing safe space launches, where

⁴ NCEI, [U.S. Billion-Dollar Weather and Climate Disasters](#), 2025

America is the world leader, it is critical to have reliable forecasts of temperature and wind extremes, or lightning conditions.

AI can reduce losses due to weather disasters. In September 2024 Helene rapidly intensified to a Category 4 hurricane and left a path of destruction 500 miles long across Florida, Georgia, North Carolina, South Carolina, Virginia, and Tennessee, killing >230 people, causing over \$50Billion of damage. Historical improvements in forecasting just from 2007-2020, which enable earlier and more targeted predictions and preparations, are estimated to reduce each hurricane's cost by 19% on average, about \$5 billion per hurricane.⁵ We need further weather forecasting improvements to track hurricanes' paths and intensity earlier and more accurately, to save lives and avoid damage to unprepared assets. AI has shown it can predict hurricane paths more accurately, forecasting that Hurricane Beryl would hit Texas not Mexico (as traditional NWP models predicted) 4 days out⁶, providing valuable extra time to prepare to save lives and reduce damage.

AI Weather Prediction (AIWP)

AI has brought a revolution in weather, and AI Weather Prediction (AIWP) is now faster, cheaper and more accurate than the traditional physics-based Numerical Weather Prediction. It is the biggest improvement in weather forecasting since satellites and supercomputers in the 1960s. The core of traditional weather forecasting is a technology called “numerical weather prediction” (NWP), which forms the foundation of nearly all weather products and services provided by both the government and private companies. Continuously developed since the 1950s, NWP models encode the physical and fluid dynamical equations which describe how our atmosphere works, and numerically solve them to predict how the weather will evolve. There has been steady progress in improving forecast skill such that today's 4-day forecast is as accurate as the next-day forecast 30 years ago⁷. These improvements in forecast skill translate into hundreds of billions of dollars saved annually through improved industry resilience to hazardous weather, and value for national security interests operating globally. Still, we want to make better decisions from weather predictions which are more accurate, more confident, and provide skill at longer lead times. For example, uncertainty in hurricane intensity and track on the Florida coast can restrict the window available for local governments to mobilize evacuations and prepare their emergency responses to just 1-2 days ahead of landfall, which makes large evacuations challenging.

⁵ Molin and Rudik, [The Social Value of Hurricane Forecasts](#), 2023

⁶ New York Times, [Artificial Intelligence Gives Weather Forecasters a New Edge](#), 2024

⁷ Our World in Data, [Weather forecasts have become much more accurate; we now need to make them available to everyone](#), 2024

NWP utilizes “data assimilation” (DA) to process observations from satellites, aircraft, weather balloons, and ground observation stations into a comprehensive state of the atmosphere. In addition to launching its own observation platforms and leveraging ones operated by other federal agencies (in particular, NASA and the DoD), NOAA acquires data for assimilation from private vendors through a Commercial Data Program⁸ (CDP). Over the past few years, tens of millions of dollars has been invested by NOAA into American companies to commercialize and deploy satellites and other platforms for weather observation, creating jobs across the country, and invigorating a competitive commercial weather observation industry.

USA contributes ~25% of the meteorological satellite data that is shared by countries globally through the World Meteorological Organization (WMO), 3% of globally-shared land surface meteorological observations and 12% of upper air radiosonde profiles⁹. All countries rely on sharing others’ observations to know what weather is coming over the horizon. Not all of these data are leveraged for traditional NWP-based forecasting. For example, many satellite photos have clouds in them, and most of these are rejected for use by traditional DA and NWP. In contrast, the AI community is rapidly developing new techniques which will benefit from these observations and use them more efficiently. In addition, the USA has large and innovative commercial observation and data technology industries. American leadership in observational data availability and capability can drive AIWP leadership.

One important way that NWP and DA are employed is to create historical replays of the weather called “reanalysis” – large (a few petabytes), 3D “video-like” datasets of dozens of important weather variables through the depth of the atmosphere. While NOAA has produced such datasets in the past, the most commonly used such dataset available today is the ECMWF Reanalysis v5 (ERA5). Reanalysis datasets are invaluable for calibrating weather forecast models; they are widely employed by the private sector to develop statistical or AI-based corrections to improve forecast skill, alongside real-world observations. But ERA5 highlights a concerning trend in the weather modeling world: that the costs to produce and run NWP systems as we chase ever-increasing forecast skill and model resolution are steeply, exponentially, and unsustainably increasing¹⁰.

On top of its immense business and research value, ERA5 produced an unintended win. By compressing decades worth of details about the atmosphere into a video-like format, it presented to the AI community a unique dataset ideally suited for building a new type of AI-based weather forecast model. Previously, AI models had been used in limited ways for “nowcasting” or extrapolation of radar and satellite imagery. They

⁸ NOAA Office of Space Commerce, [NOAA Commercial Data Program](#)

⁹ Phys.org, [US 'vital' for forecasting global weather extremes: UN](#), 2025

¹⁰ ECMWF, [What next for supercomputing in weather forecasting?](#), 2016

lacked an adequate training dataset to tackle NWP-like weather forecasting. ERA5 – which provided a high-quality and complete replay of the atmosphere – simplified the forecast problem into an AI engineering one: Given the last two “frames” of weather, can we predict the next one? Rolling this process forward in time by feeding the model’s predictions of the next timestep back into itself produces a forecast.

Between 2018 and 2021, a few meteorologist-led efforts produced proofs of concept that AI models trained on ERA5 could emulate NWP forecasts. But in 2022, efforts from outside the weather community – by experienced AI model developers and AI research teams at large tech companies – started producing significant research advances. By 2023, AIWP research models achieved (and sometimes outperformed) forecast skill of comparable, best-in-class NWP systems operated by NOAA and ECMWF. This technology has proliferated, with a vibrant commercial ecosystem building new AIWP research models. Almost all of these AIWP models rely on European re-analysis data - the ERA5 dataset. The leading operational AIWP model available today is ECMWF’s AIFS.

AIWP promises not only a new way to forecast the weather, but also opens a new frontier for improving forecast accuracy. AIWP models are up to 10,000x faster to run inference (produce a forecast) while consuming a trivial amount of resources – one commodity GPU, rather than a large-scale High Performance Computing (HPC) super-computer system using tens of thousands of cores. With these tools, forecasters anticipating a devastating tropical cyclone blow to Florida can simulate hundreds or thousands of weather scenarios in just minutes, for about the price of a cup of coffee. And although training these models for this sort of task is costly, it is still several orders of magnitude cheaper than doing so for frontier Large Language Models (LLMs) and applications such as ChatGPT or DeepSeek.

The next evolution in AIWP is forecasting directly from “raw” weather observations, bypassing traditional NWP and DA systems altogether. This will leverage orders of magnitude more observational data than traditional NWP, which only uses a small subset of data that has been assimilated with expensive DA. Such an AIWP system – powered by the federal government’s datastream of weather observations – will forecast faster and more frequently, providing more rapid updates to emerging hazardous weather scenarios and serving both public safety and efficient business operations. Furthermore, this innovation will open up NOAA’s massive, petascale archive of historical weather data for training direct modeling applications. Forecasting directly from observations will bring unprecedented accuracy and utility – a vital tool for securing economic resilience in the face of extreme weather domestically and around the globe.

Recommendations

Through its AI Action Plan, the White House should coordinate an “AI Weather Prediction Moonshot” centered around two key areas - AIWP modeling and supporting observations data. The goal is American global weather forecasting leadership, with a new capability to produce the most accurate forecasts for the oceans and atmosphere anywhere in the world.

1. NOAA should deliver an operational AIWP model for global medium-term forecasting, with performance competitive to the ECMWF’s AIFS, by July 4th 2026.
 - a. Europe’s AI forecasting model went operational in February 2025 (ECMWF’s AIFS) after a year-and-a-half of development by a team of ~ 15 people, iterating on leading technology from the private sector (Google DeepMind’s Graphcast).
 - b. China has a number of AIWP models and the China Meteorological Administration is moving to put them in operation: Pangu-Weather and Zhiji weather (Huawei), Baguan (Alibaba), FuXi (Shanghai Academy of Artificial Intelligence for Science).
 - c. American technology companies have produced a variety of leading AIWP models, including GraphCast / GenCast (Google DeepMind), Aurora (Microsoft), and FourCastNet / StormCast (NVIDIA). A vibrant technology startup community is regularly producing competitive research models.
 - d. Within NOAA, efforts with limited resources are underway to attempt to build internal AIWP capacity for both global (GraphCast-GFS) and regional (WoFSCast) applications for early evaluation, but these are only small efforts to develop limited new in-house capabilities, likely to take years not months.
2. NOAA should source the latest AIWP/data technology from private industry partners via acquisition, licensing, or co-development.
 - a. Establish a new NOAA AI Strategic Plan to achieve leadership in AIWP. NOAA’s pre-AIWP AI Strategic Plan expires in 2025. Efforts in AIWP need to accelerate to catch up with Europe and China, and benefit from the latest technology in private companies.
 - b. Increase efficiency of AIWP modeling activities across the federal government by centralizing them in a single line office within NOAA. There are currently multiple AIWP R&D projects across the DoD, DOE, NOAA, and NASA, which fragments resources.
 - c. Contract AIWP development from the private sector. The staggering pace of AIWP development is extremely difficult to replicate within federal

agencies. Instead, NOAA and stakeholders/partners should prioritize working with the leading AIWP technology players, many of whom exist outside of the traditional “weather enterprise.” These players are likely better resourced for agile technology development, but more importantly bring in critical AI expertise and leadership poorly represented in the meteorology community.

- d. Use new types of public-private partnerships for developing AIWP technology. Traditional collaboration mechanisms (Small Business Innovation Research grants and Collaborative Research and Development Agreements - SBIRs and CRADAs) provide insufficient incentives for private companies to quickly build AIWP models in collaboration with NOAA. NOAA needs to be able to quickly allocate \$Tens-of-Millions in budget to private partners developing the latest AI and data technologies. The Commercial Data Program (CDP) has shown some success here, aligning timescales, funding, and IP ownership with the needs of private technology companies, to drive innovation.
3. NOAA, NASA, DoD, DoE, and other agencies should coordinate and leverage the USA’s unparalleled Earth Observation capabilities to provide the world’s leading real-time datastream of weather observations for AIWP modeling.
- As AIWP moves to forecasting directly from observations, the quantity, quality, and real-time availability of weather observation data will become a key competitive advantage for both training models as well as deploying and running them operationally
 - The USA produces leading observational datasets and systems, and invests in an increasingly innovative commercial observation industry through NOAA’s CDP. These datasets should be built to directly support the development of AIWP technology.
 - NOAA and other agencies should deliver the lowest-latency, most reliable datastream of weather observations for AIWP innovation by continuing to convert datasets to analysis-ready/cloud-optimized and AI-ready formats and by bolstering capabilities to disseminate data in real-time, rather than tied to the operational forecast modeling cycle.

Conclusion

Becoming the global leader in AI for weather forecasting is a key pillar of enhancing America’s broader AI leadership. Being the global leader in AI weather would further enhance our leadership in agriculture, energy, logistics, space, and in military theaters around the world.

The key to establishing and sustaining this leadership is to support vibrant public-private collaborations, closely working with a highly efficient and AI mission-driven NOAA. In addition to bolstering its modeling capabilities with AI, NOAA should deliver and make available the leading real-time operational weather observation data pipeline, which will drive further investment in novel and impactful forecast models by their private-sector counterparts. NOAA should contract with one or more American companies to build and deliver an operational and best-in-class global medium-term AI Weather Prediction Model (with competitive results to ECMWF's AIFS model) in 12 months.

When technology moves very fast in an area, as it is in AI Weather, it is important to tap leading-edge AI talent, often in startups and tech companies, or risk being left behind.

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